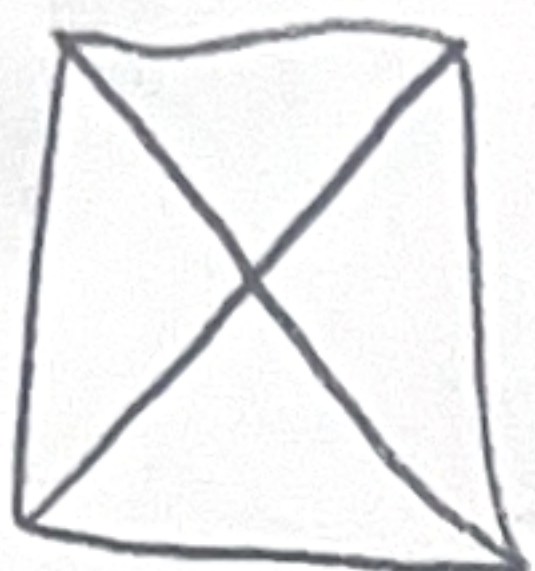
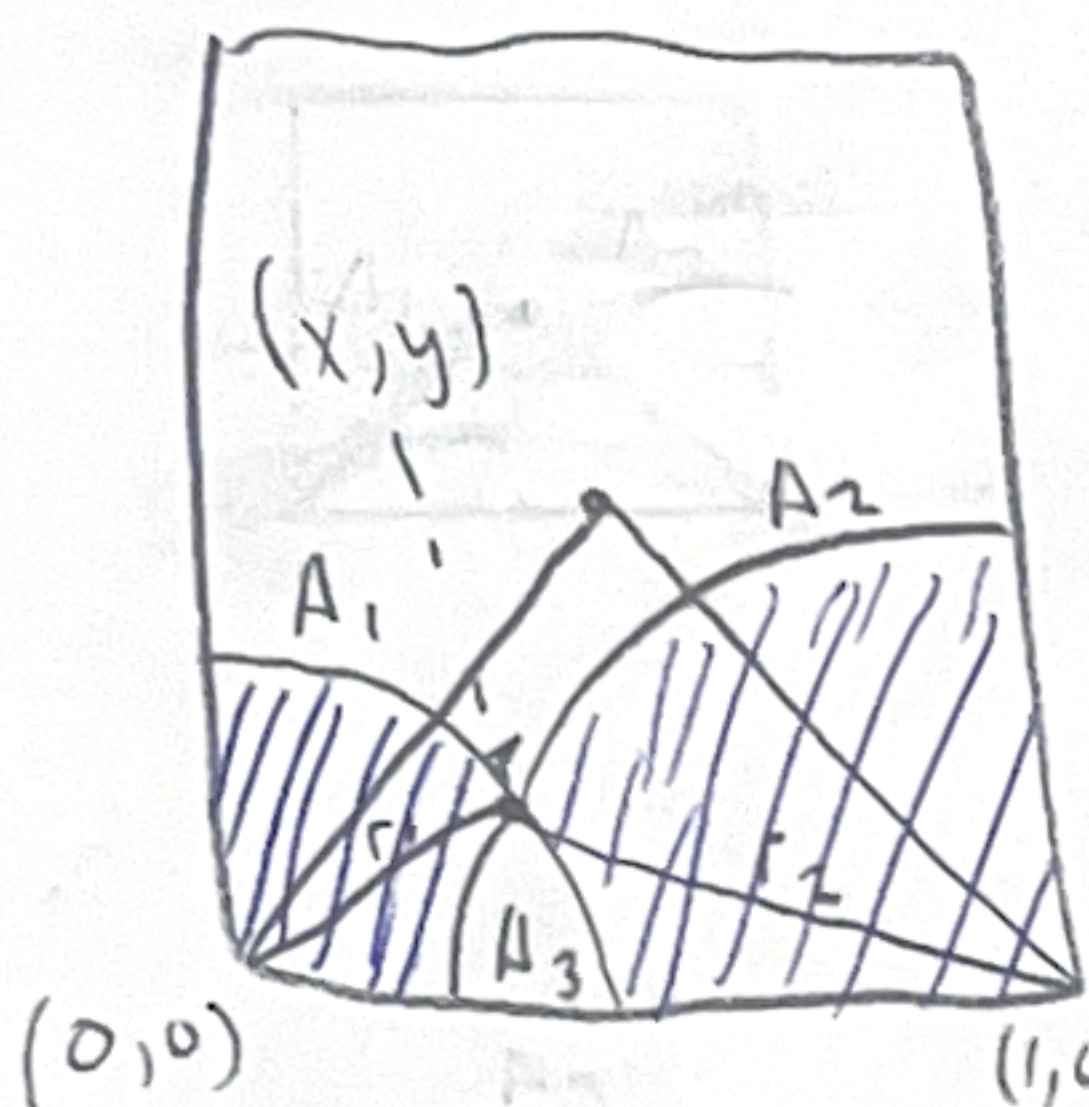


Besides The Point

Q: Two random points, one red and one blue, are chosen uniformly and independently from the interior of a square. To ten decimal places, what is the probability that there exists a point on the side of the square closest to the blue point that is equidistant to both the blue point and the red point?

A:  WLOG, each triangle section is equally probable, due to symmetry of the square.

 For any given point (x,y) , the shaded region is the probability of success. Given this we need to integrate over the triangle and solve for $A_1 + A_2 - 2A_3$

$$r_1 = \sqrt{x^2 + y^2}$$

$$r_2 = \sqrt{(x-1)^2 + y^2}$$

$$A_1 = \pi(x^2 + y^2) \quad A_2 = \pi((x-1)^2 + y^2)$$

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$$A_1 + A_2 = \pi(x^2 + y^2 + (x-1)^2 + y^2) = \pi(2x^2 + 2y^2 - 2x + 1)$$

$$\iint A_1 + A_2 dy dx = \int_0^{1/2} \int_0^{1/2 - |x - 1/2|} A_1 + A_2 dy dx = \int_0^{1/2} \int_0^x A_1 + A_2 dy dx + \int_{1/2}^1 \int_0^{1-x} A_1 + A_2 dy dx$$

$$\int_0^{1/2} \int_0^x \pi(2x^2 + 2y^2 - 2x + 1) dy dx + \int_{1/2}^1 \int_0^{1-x} \pi(2x^2 + 2y^2 - 2x + 1) dy dx$$

$$\pi \left[\int_{1/2}^1 \int_0^{1-x} 2x^2 + 2y^2 - 2x + 1 dy dx + \int_0^{1/2} \int_0^x 2x^2 + 2y^2 - 2x + 1 dy dx \right]$$

$$\pi \left[\int_0^{1/2} \left[2x^2y + \frac{2y^3}{3} - 2xy + y \right]_0^x dx + \int_{1/2}^1 \left[2x^2y + \frac{2y^3}{3} - 2xy + y \right]_0^{-x+1} dx \right]$$

$$\pi \left[\int_0^{1/2} 2x^3 + \frac{2x^3}{3} - 2x^2 + x dx + \int_{1/2}^1 2x^2(-x+1) + \frac{2}{3}(-x+1)^3 - 2x(-x+1) - x+1 dx \right]$$

$$\pi \left[\int_0^{1/2} 2x^3 + \frac{2x^3}{3} - 2x^2 + x dx + \int_{1/2}^1 -\frac{8}{3}x^3 + 6x^2 - 5x + \frac{5}{3} dx \right]$$

$$\pi \left[\left(\frac{x^4}{2} + \frac{x^4}{6} - \frac{2x^3}{3} + \frac{x^2}{2} \right) \Big|_0^{1/2} + \left(-\frac{2}{3}x^4 + 2x^3 - \frac{5x^2}{2} + \frac{5}{3}x \right) \Big|_{1/2}^1 \right]$$

$$\pi \left[\left(\frac{1}{32} + \frac{1}{96} - \frac{1}{12} + \frac{1}{8} \right) + \left(\left(-\frac{2}{3} + 2 - \frac{5}{2} + \frac{5}{3} \right) - \left(-\frac{1}{24} + \frac{1}{4} - \frac{5}{8} + \frac{5}{6} \right) \right) \right]$$

$$\pi \left[\frac{1}{12} + \left(\frac{1}{12} - \frac{5}{12} \right) \right]$$

$$\pi \frac{2}{12} = \frac{\pi}{6} = A_1 + A_2$$

$$A_3 = \text{overlap} = 2 \left[\underbrace{r_1^2 \cos^{-1} \left(\frac{1+r_1^2-r_2^2}{2r_1} \right)}_{\text{I}} + \underbrace{r_2^2 \cos^{-1} \left(\frac{1+r_2^2-r_1^2}{2r_2} \right)}_{\text{II}} - \underbrace{\frac{1}{2} \left[(-1+r_1+r_2)(1+r_1-r_2)(1+r_1+r_2)(1+r_1-r_2) \right]}_{\text{III}} \right]$$

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For

$$\text{I: } r_1^2 \cos^{-1} \left(\frac{1+r_1^2-r_2^2}{2r_1} \right) \text{ substituting for } r_1 \text{ and } r_2:$$

$$(x^2+y^2) \left(\cos^{-1} \left(\frac{1+(x^2+y^2)-(x-1)^2+y^2}{2\sqrt{x^2+y^2}} \right) \right) = (x^2+y^2) \left(\cos^{-1} \left(\frac{2x}{2\sqrt{x^2+y^2}} \right) \right) =$$

$$\text{I} = (x^2+y^2) \left(\cos^{-1} \left(\frac{x}{\sqrt{x^2+y^2}} \right) \right) = (x^2+y^2) \cos^{-1} \left(\frac{x}{\sqrt{x^2+y^2}} \right)$$

$$\text{II: } r_2^2 \cos^{-1} \left(\frac{1+r_2^2-r_1^2}{2r_2} \right) \text{ substituting for } r_1 \text{ and } r_2:$$

$$((x-1)^2+y^2) \cos^{-1} \left(\frac{1+((x-1)^2+y^2)-(x^2+y^2)}{2\sqrt{(x-1)^2+y^2}} \right) = ((x-1)^2+y^2) \cos^{-1} \left(\frac{-2+2x}{2\sqrt{(x-1)^2+y^2}} \right)$$

$$\text{II} = ((x-1)^2+y^2) \cos^{-1} \left(\frac{1-x}{\sqrt{(x-1)^2+y^2}} \right)$$

$$III: \frac{1}{2} \sqrt{(-1+r_1+r_2)(1+r_1-r_2)(1-r_1+r_2)(1+r_1+r_2)}$$

Substitute r_1 and r_2

$$A_1 + A_2 = \frac{\pi}{6}$$

$$\frac{1}{2}(-1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} - \sqrt{(x-1)^2+y^2})(1 - \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2}) = III$$

$$A_3 = 2(III + I - I)$$

$$A_3 = 2 \left[(x^2+y^2) \cos^{-1}\left(\frac{x}{\sqrt{x^2+y^2}}\right) + ((x-1)^2+y^2) \cos^{-1}\left(\frac{1-x}{\sqrt{(x-1)^2+y^2}}\right) - \frac{1}{2}(-1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} - \sqrt{(x-1)^2+y^2}) \cdot (1 - \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2}) \right]$$

$2A_3$ integral:

$$= \int_0^1 \int_0^{1/2-|x-1/2|} 2A_3 \, dy \, dx$$

$$= 4 \int_0^1 \int_0^{1/2-|x-1/2|} \left[x^2+y^2 \cos^{-1}\left(\frac{x}{\sqrt{x^2+y^2}}\right) + ((x-1)^2+y^2) \cos^{-1}\left(\frac{1-x}{\sqrt{(x-1)^2+y^2}}\right) - \frac{1}{2}(-1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} - \sqrt{(x-1)^2+y^2}) \cdot (1 - \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2})(1 + \sqrt{x^2+y^2} + \sqrt{(x-1)^2+y^2}) \right] dy \, dx$$

Solved using Matlab

$$2A_3 \text{ integral} \approx 4 \times 0.008047799189990 \approx 0.032191196759960$$

Back to $P(\text{Success})$ is equal to $A_1 + A_2 - 2A_3$ integrated over triangle region

$$A_1 + A_2 - 2A_3$$

$$= \frac{\pi}{6} - 0.032191196759960$$

$$= .491407578838339$$

To ten decimal places:

$$\approx 0.4914075788$$